Mostafa Elshabasy

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Professional Summary

CS undergraduate with experience in ***applied ML/AI research*** and ***aerospace systems engineering***. Trained and benchmarked small language models on GPU clusters using PyTorch and Hugging Face; built simulation tools for orbital mechanics and spacecraft integration testing. Proficient in Python, React, Node.js, and AWS – directly aligned with NASA Ames' research in data science, simulation, and airspace operations.

AWARDS AND CERTIFICATIONS

Presidential Academic Scholarship | *The George Washington University* — *2022 – Present*
Dean's List (SEAS) | *The George Washington University* — *Spring 2023*
AWS Academy Graduate: Cloud Foundations | *AWS Academy* — *February 2025*
JIS Higher Honor Roll | *Jubail International School* — *Fall 2019 – Winter 2022*
EAI MobiQuitous 2025 – Presented "Tiny Models, Tough Limits" | *Shanghai, China (Online)* — *November 2025*

Technical Skills

Languages: Python, JavaScript, C, C++, Java, Kotlin, SQL, HTML/CSS, Verilog HDL
AI/ML: PyTorch, Hugging Face Transformers, Generative AI, model evaluation (accuracy/F1/latency/robustness)
Cloud: AWS (EC2, Lambda, S3, RDS, DynamoDB, API Gateway), cloud-based data pipelines
Web/Viz: React, Node.js, Flask, Bootstrap, MySQL, Three.js (3D), Canvas API
Tools: Linux, Git, Quartus Prime, Icarus Verilog, HIL/SIL Testing, CI-style workflows

Experience

Software Integration & Testing Intern | *Astrobotic Technology (NASA CLPS)* — *Pittsburgh, PA Aug 2025 – May 2026*
- Built and maintained ***HIL*** and ***SIL*** test automation for flight/avionics interfaces; expanded the HSS emulator with new protocol features and refactored 5+ emulators, reducing debug time by ∼30%.
- Updated DFS & EFS systems, improving reliability and increasing test coverage by ∼25%.; collaborated cross-functionally across firmware and avionics teams.

AI Research Intern (REU: ASSET Program) | *National Science Foundation (NSF)* — *Miami, FL May 2025 – Aug 2025*
- Trained and fine-tuned 5+ small language models (DistilBERT, MobileBERT, TinyBERT, Phi-3 Mini, TinyLLaMA) on FIU's Diamond GPU cluster using Python, PyTorch, and Hugging Face Transformers.
- Built data engineering pipelines to benchmark models across 6+ metrics (accuracy, F1, latency, robustness), producing consolidated JSON/CSV reports for analysis and visualization.
- Collaborated with faculty PI, postdocs, and PhD students; presented at EAI MobiQuitous 2025 – "Tiny Models, Tough Limits."

Web Assistant | *Institute for International Economic Policy (IIEP)* — *Washington, DC Aug 2024 – Present*
- Maintained 80%+ of IIEP's website (Drupal/HTML/CSS); deployed 30+ articles monthly and improved navigation, contributing to a 15% drop in bounce rate.

Instructor – Coding/STEM for Kids | *Codeadvantage* — *Washington, DC Dec 2024 – Present*
- Taught programming concepts (loops, conditionals, event-driven logic) to 10+ students via Minecraft: Education Edition and LEGO robotics.

Education

The George Washington University — **B.S. in Computer Science | GPA: 3.60 / 4.0** | Washington, DC — *Expected May 2027*

Projects

Kessler | *Three.js, SGP4, CelesTrak TLE Data, JavaScript* — *2025*
- Real-time 3D browser simulation of orbital debris (Kessler Syndrome) using live TLE data and SGP4 propagation – the same standard used by NASA for satellite tracking. Zero backend, zero dependencies.

Codemedic | *C++17, libclang, Anthropic API, CMake* — *2025*
- Built a C/C++ compiler error fixer that isolates root-cause errors via dependency graph, synthesizes LLM patches, and recompiles to verify; supports batch mode, git auto-commit, and custom compiler flags.

Porkchop Plotter | *JavaScript, Lambert Solver, Keplerian Ephemeris, Canvas API* — *2025*
- Mission-grade interplanetary trajectory tool using a Lambert arc solver. Renders interactive C3 energy porkchop plots with optimal launch window detection; zero dependencies.